

COURSEWORK
FINANCIAL ANALYTICS AND MACHINE LEARNING
(PERSONAL HOMEWORK)

INSTRUCTIONS

Each student is required to submit a short report on one of the suggested papers in PDF format through the dedicated submission point. The report must be at most **3 pages long**, excluding references and appendices, if any.

The goal of the assignment is to produce a concise but self-contained report on the selected paper. The report should include the following elements.

- (1) **Description of the paper.** Provide a clear summary of the problem addressed in the paper, the machine learning methodology used by the authors, the proposed solution, and the main empirical results obtained in the original study.
- (2) **Python implementation and empirical replication.** Try to reproduce the methodology of the paper in Python and apply it to the available dataset in order to obtain results that are broadly comparable to those reported in the paper. It is not mandatory to reproduce the exact numerical results of the paper, since this may be difficult when the available data are not exactly the same as those used by the authors. The key requirement is that the replication is methodologically sound, correctly implemented, and clearly explained. The report should describe the main steps of the implementation and may include screenshots or excerpts of the most relevant parts of the code.
- (3) **Presentation of results.** Produce graphs and/or tables summarizing the main results obtained from your implementation. It is not necessary to reproduce all the empirical results of the paper. Instead, you should focus on the results that you consider most important and explain why these results were selected.
- (4) **Critical comparison and discussion.** Compare your findings with those reported in the original paper. Discuss similarities and differences, possible reasons for any discrepancies, and the main insights emerging from your replication. The discussion should also explain why these insights are relevant from a financial, econometric, or machine learning perspective.

EVALUATION CRITERIA

The assignment will be graded according to the following criteria:

- **Understanding of the paper and of the financial problem:** clarity in explaining the research question, the financial motivation, the proposed machine learning approach, and the main findings of the original paper.
- **Correctness of the Python implementation:** consistency between the implemented methodology and the paper, appropriate preprocessing, correct model training and validation, and transparent coding choices.
- **Quality and robustness of the empirical replication:** ability to obtain meaningful results using the available data, even if exact numerical replication is not achieved; attention to robustness, limitations, and possible sources of discrepancy.
- **Quality of tables, figures, and result presentation:** clarity, relevance, and interpretability of the empirical evidence produced by the students.
- **Critical comparison with the original paper:** ability to compare the obtained results with those of the authors, discuss differences, and extract relevant insights.
- **Clarity, organization, and conciseness:** structure of the report, quality of writing, and compliance with the page limit.